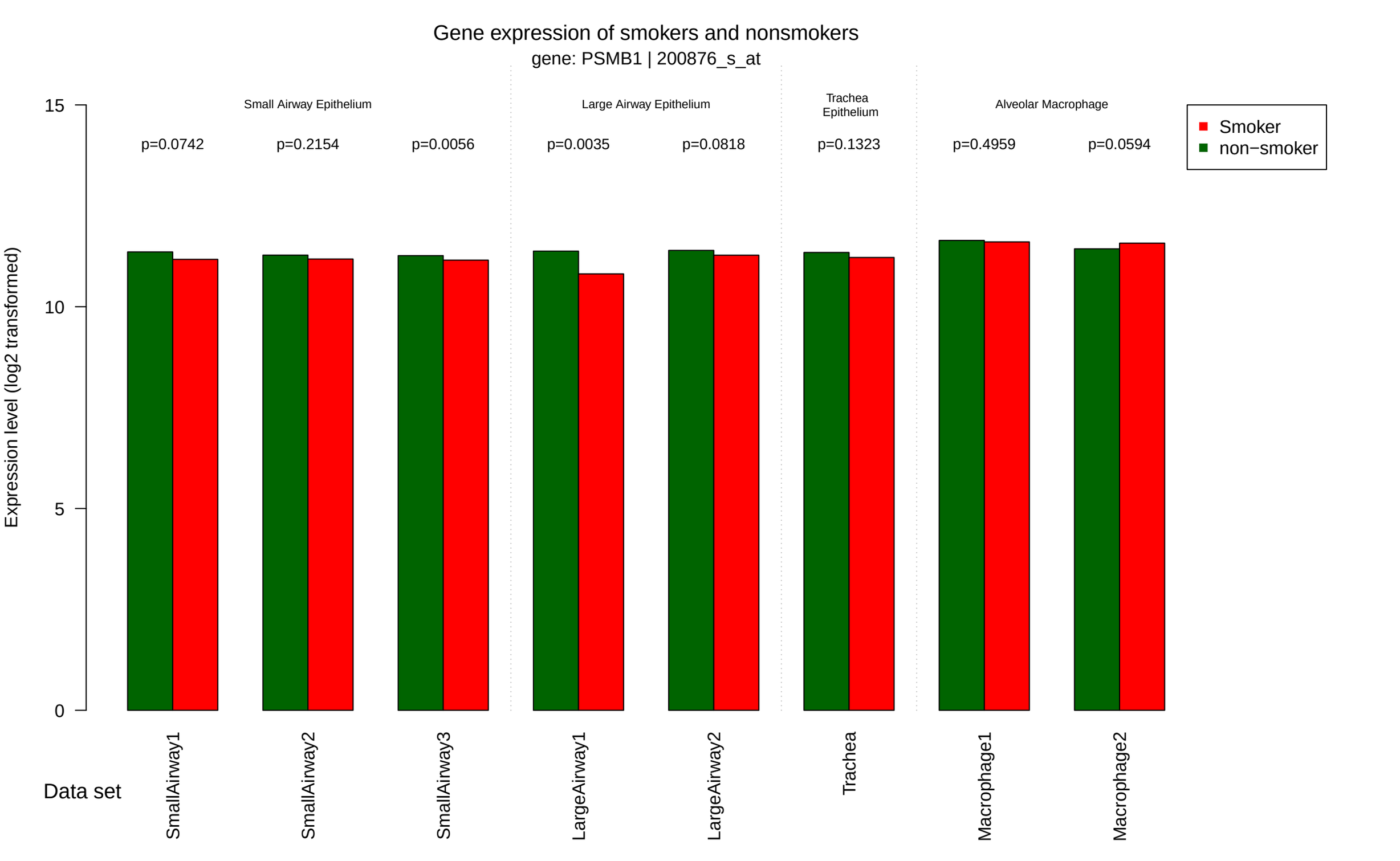
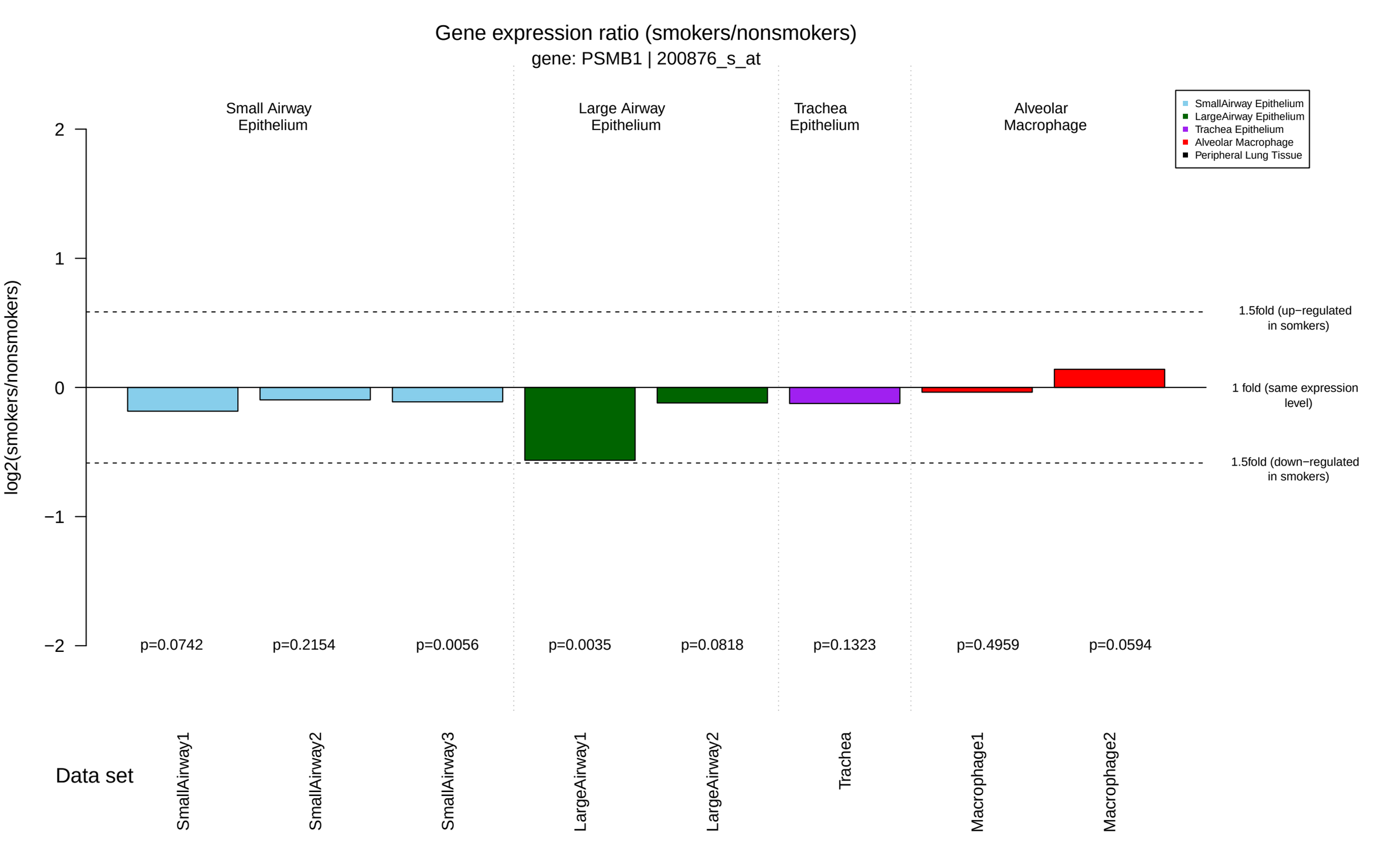
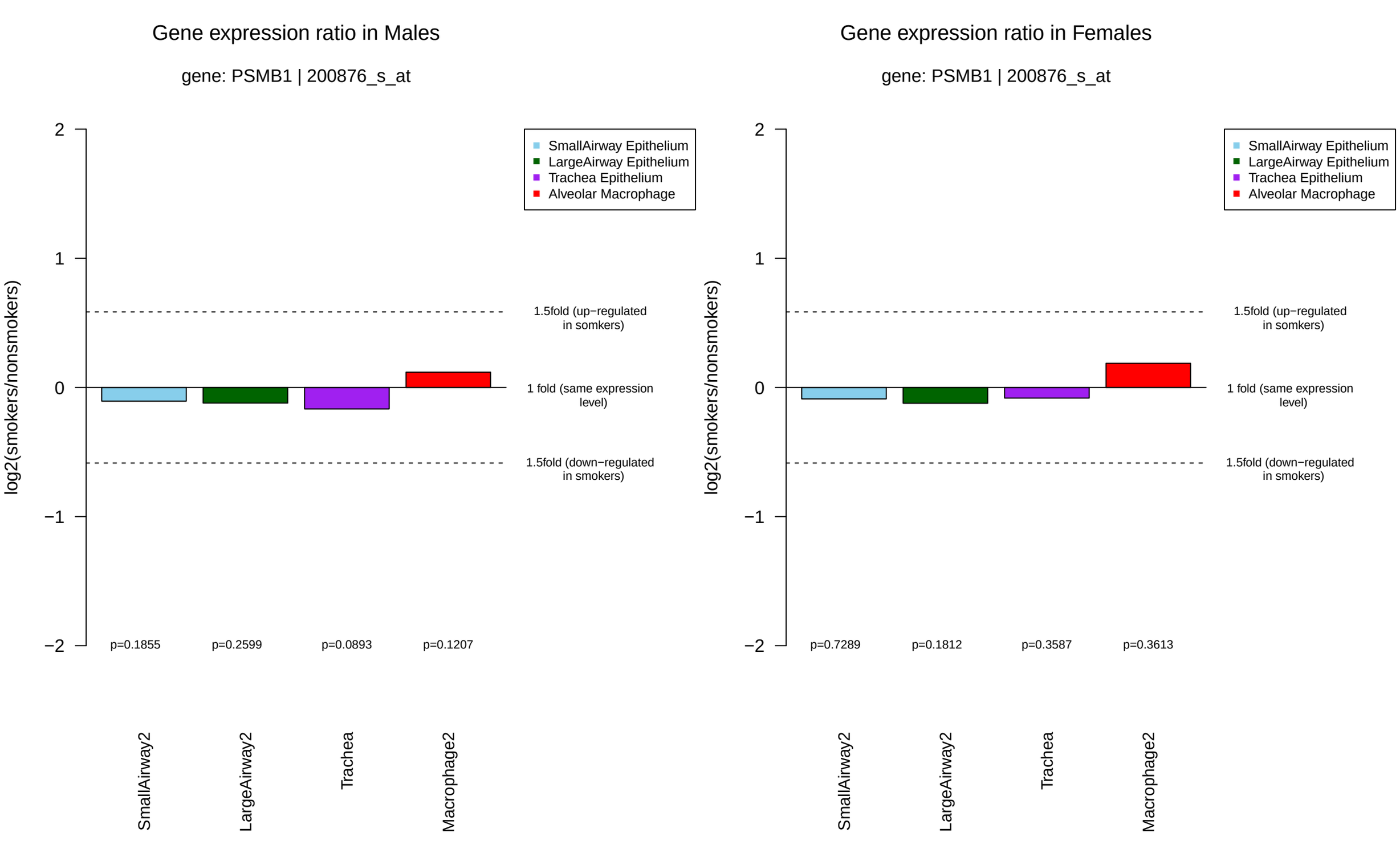
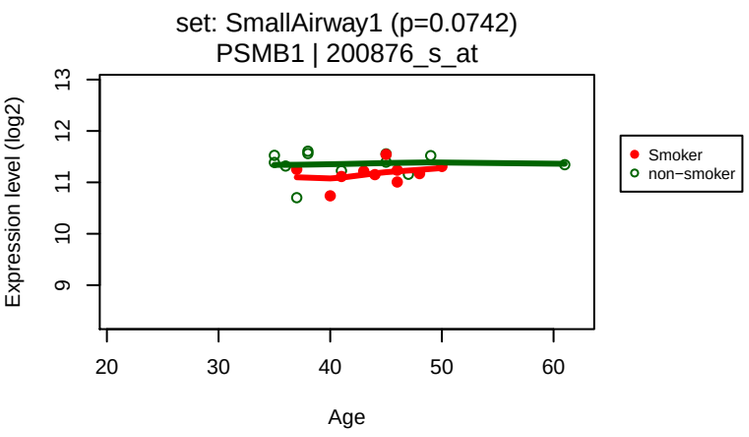
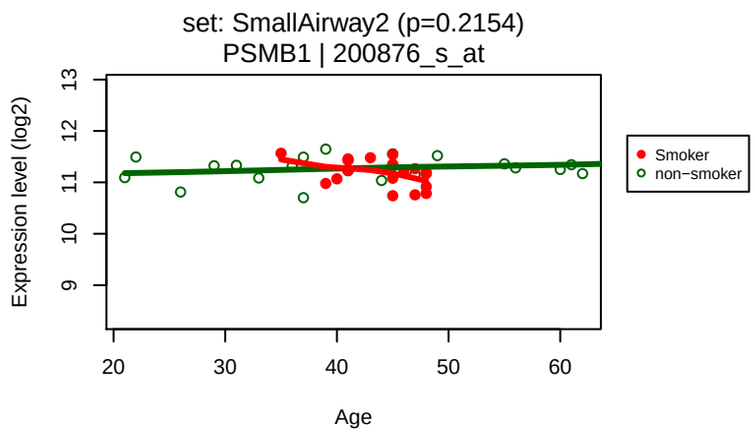
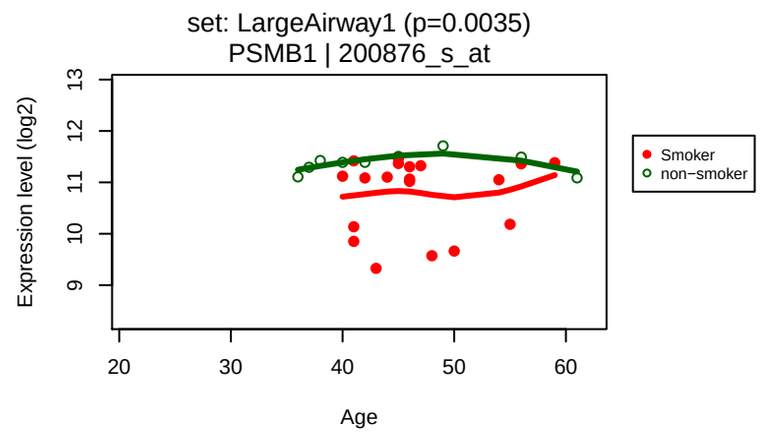

Gene expression of smokers and nonsmokers
gene: PSMB1 | 200876_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0742
p=0.2154
p=0.0056
p=0.0035
p=0.0818
p=0.1323
p=0.4959
p=0.0594
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

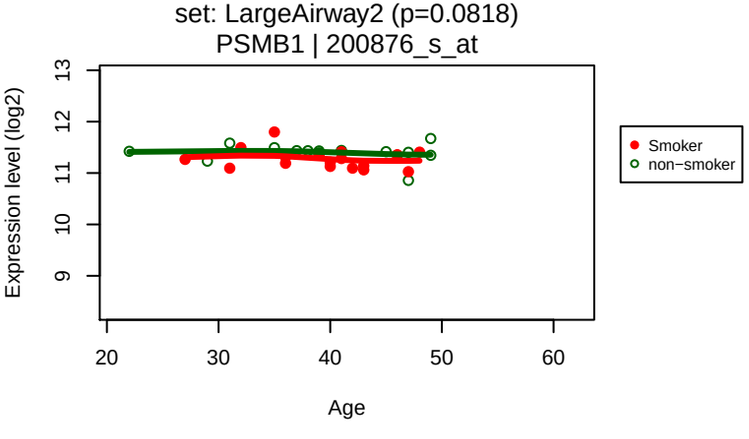

Gene expression ratio (smokers/nonsmokers)
gene: PSMB1 | 200876_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0742
p=0.2154
p=0.0056
p=0.0035
p=0.0818
p=0.1323
p=0.4959
p=0.0594
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

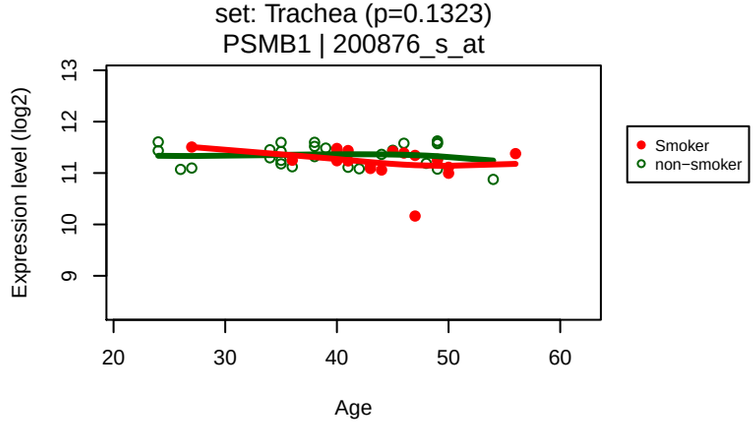

Gene expression ratio in Males
gene: PSMB1 | 200876_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1855
p=0.2599
p=0.0893
p=0.1207
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMB1 | 200876_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7289
p=0.1812
p=0.3587
p=0.3613
SmallAirway2
LargeAirway2
Trachea
Macrophage2

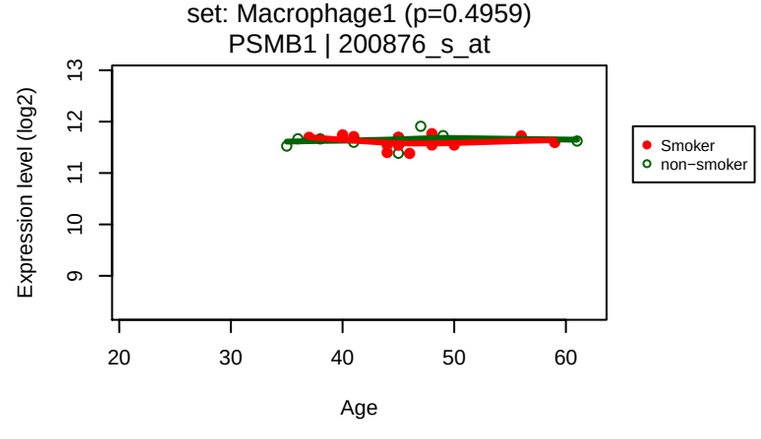

set: SmallAirway1 (p=0.0742)
PSMB1 | 200876_s_at
Expression level (log2)
Age
Smoker
non−smoker

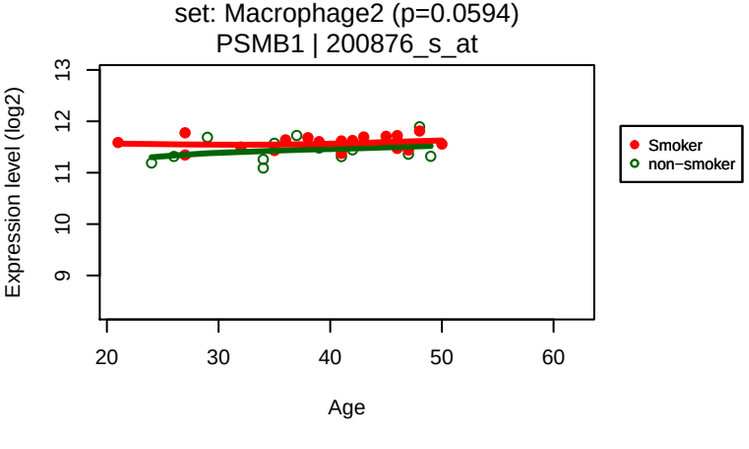

set: SmallAirway2 (p=0.2154)
PSMB1 | 200876_s_at
Expression level (log2)
Age
Smoker
non−smoker

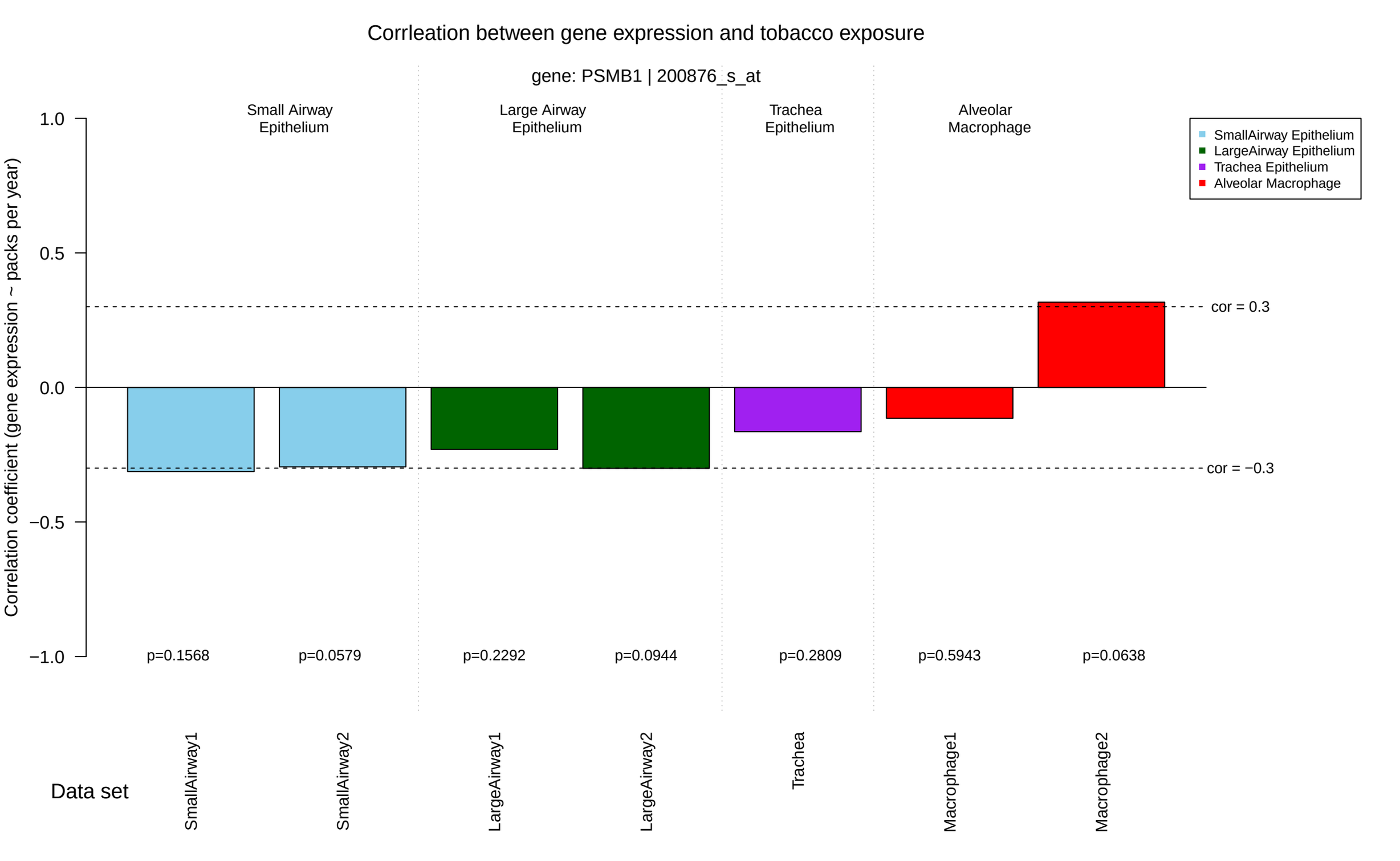

set: LargeAirway1 (p=0.0035)
PSMB1 | 200876_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0818)
PSMB1 | 200876_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1323)
PSMB1 | 200876_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4959)
PSMB1 | 200876_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0594)
PSMB1 | 200876_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMB1 | 200876_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1568
p=0.0579
p=0.2292
p=0.0944
p=0.2809
p=0.5943
p=0.0638
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2